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Roll No. 811241302 8

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BE (7th Semester)
Examination, Nov.-Dec., 2016

(New Scheme)

Electrical Drives

Time Allowed : 3 hours *Maximum Marks :* 80
Minimum Pass Marks : 28

Note : (*i*) Attempt **all** questions. Part (*a*) of each question is compulsory. Attempt any **two** parts from (*b*), (c) and (*d*).

(*ii*) The figures in the right-hand margin indicate marks.

1. (*a*) Define Active load torque and Passive load torque. Give suitable examples. [2]

(*b*) Describe the four quadrant operation of a drive. Draw neat diagram and give suitable example. [7]

(*c*) What is load equalisation? Derive expression for mechanical time constant of a motor having a flywheel mounted on its shaft. [7]

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(*d*) A horizontal conveyor belt moving at uniform speed of 1.2 m/s transports material at the rate of 100 tons/hr. Belt is 200 m long and driven by a motor at 1200 rpm.

(*i*) Determine the load inertia referred to motor shaft.

(*ii*) Calculate the torque that motor should develop to accelerate the belt from standstill to full speeds in 8 second. Moment of inertia of motor is 0.1 kg-m^2. [7]

2. (*a*) Differentiate between Variable speed and Multi speed drive. [2]

(*b*) Describe closed loop operation of multi-motor drive. Explain the significance of progressive draw arrangement. Give suitable example. [7]

(*c*) Derive an expression for overloading factor for a short time duty motor. [7]

(*d*) Half hour rating of a motor is 100 kW. Heating time constant is 80 minutes and the maximum efficiency occurs at 70% full load. Determine continuous rating of motor. [7]

3. (*a*) Write two basic requirements of a DC motor starter. [2]

(*b*) Draw the dynamic equivalent circuit of DC motor and derive expressions for transient analysis of separately excited motor with armature control. [7]

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(*c*) Describe the principle of operation of single phase fully controlled rectifier control of DC separately excited motor. [7]

(*d*) A 30 kW, 220 DC shunt motor with a full load speed of 535 rpm is to be braked by plugging. Estimate the value of resistance which should be placed in series with it to limit the initial current to 200A. What would be the value of initial braking torque? Also find the value of braking torque when the speed had fallen to half of its full load value? [7]

4. (*a*) Write two applications of damper winding in a Synchronous motor. [2]

(*b*) Describe the working principle of a voltage source inverter controlled induction motor drive. Draw suitable diagram. [7]

(*c*) Explain the principle of operation of static Scherbius drive. Draw suitable diagram. [7]

(*d*) Describe self-controlled synchronous motor drive employing load commutated thyristor inverter. [7]

5. (*a*) What is dynamic braking? [2]

(*b*) For good adhesion the motor speed torque characteristic should have low speed regulation and for good sharing of loads between motors, the speed torque curve should have high speed regulation. Justify. [7]

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(*c*) Derive the expression for the total tractive effect required to move the train. Explain the notations used. [7]

(*d*) An electric train weighing 500 Tonnes climbs up gradient with G=8 and following speed time pattern :

(*i*) Uniform acceleration of 2.5 km/hr/sec for 60 sec

(*ii*) Constant speed for 5 min

(*iii*) Costing for 3 min

(*iv*) Dynamic braking at 3 kmphps to rest

The train resistance in 25 N/Tonne, rotational inertia effect 10% and combined efficiency of transmission and motor is 80%. Calculate specific energy consumption. [7]

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